

1. Introduction

This report describes how our Mood Prediction Tool estimates a user's general mood state from six simple survey inputs. It covers:

1. Which inputs are used and why they matter
2. The step-by-step logic: first rule-based classification, then a weighted regression, then final category thresholds
3. An example walkthrough to illustrate the process
4. Notes on limitations and ethical considerations

By following this document, any team member can understand, reproduce, or extend the model.

2. Model Inputs

Each input is easy for users to report, backed by public-health research, and linked to mental wellbeing:

Input	Description	Why It Matters	Reference
Sleep (S)	Hours slept per night (past week)	Less sleep (≤ 6 hrs) increases risk of frequent mental distress; more sleep generally improves mood	CDC: Inadequate sleep linked to mental distress (CDC)
Physical Activity (A)	Hours per week of moderate/vigorous activity	Regular exercise correlates with better mental health and reduced anxiety/depression symptoms	WHO physical activity fact sheet (Australian Bureau of Statistics)
Age (D)	Age in decades ($\text{Age} \div 10$)	Mood patterns differ across life stages; aging tends to decrease average mood slightly	ABS mental health survey methodology (Australian Bureau of Statistics)
Psychological Distress (K)	K10 score $\div 10$ (range based on 10–50 scale)	Higher K10 scores indicate greater anxiety/depression symptoms and poorer mood	ABS Users' Guide on K10 scale (Australian Bureau of Statistics)
Self-Rated Health (H)	Health scale 1 ('Poor') to 5 ('Excellent')	People rating their health as poorer tend to have lower overall wellbeing and mood	ScienceDirect overview on SRH predictive value (ScienceDirect)
Social Connectedness (C)	Number of close contacts or social hours	Greater connectedness reduces depression/anxiety risk and improves emotional resilience	PLOS One longitudinal review (PLOS)

3. Rule-Based Pre-Classification

We begin by applying simple **if–then rules** to catch extreme cases quickly:

Rule Name	Condition	Classification
Low-Mood Rule	Sleep ≤ 5 hours AND K10 ≥ 35	Mood = Low
High-Mood Rule	(Sleep ≥ 7 hours OR Activity ≥ 5 hours) AND Social contacts ≥ 3	Mood = High
Default	Any other combination	Proceed to regression

4. Regression Model: Calculating continuous MoodScore

After rule-based filtering, we compute a continuous *MoodScore* using a weighted linear regression:

Input Variable	Coefficient	Contribution Format
Sleep (<i>S</i>)	0.25	$0.25 \times S$
Activity (<i>A</i>)	0.20	$+ 0.20 \times A$
Age in decades (<i>D</i>)	− 0.15	$- 0.15 \times D$
K10 distress ($\div 10$) (<i>K</i>)	− 0.30	$- 0.30 \times K$
Self-rated Health (<i>H</i>)	0.10	$+ 0.10 \times H$
Social Contacts (<i>C</i>)	0.10	$+ 0.10 \times C$
Constant term	2.00	Base mood offset

MoodScore formula:

$$\text{MoodScore} = 0.25 \times S + 0.20 \times A - 0.15 \times D - 0.30 \times K + 0.10 \times H + 0.10 \times C + 2.00$$

- **Coefficients** reflect research-driven effect sizes:
 - Sleep contributes most positively (0.25 per hour).
 - Distress subtracts heavily (0.30 per “K10 $\div 10$ ”).
- **Scaling inputs** (dividing K10 and Age) ensures consistency of units.

5. Final Classification via Thresholds

We convert the continuous *MoodScore* (theoretical range 0–10) into three categories:

- High Mood:** MoodScore ≥ 7
- Medium Mood:** $4 < \text{MoodScore} < 7$
- Low Mood:** MoodScore ≤ 4

This three-tier system is easy for users to interpret on the dashboard.

6. Example Walk-Through

Variable	Value	Symbol
Sleep (S)	6 hrs	$S = 6$
Physical Activity (A)	4 hrs	$A = 4$
Age (D)	32 yrs	$D = 3.2$
Psychological Distress (K)	28 points	$K = 2.8$
Self-Rated Health (H)	“Good” = 4	$H = 4$
Social Connectedness (C)	3 total	$C = 3$

a) Rule Check

- Low-Mood? No ($S > 5$, $K < 3.5$)
- High-Mood? No ($C = 3$ but neither $S \geq 7$ nor $A \geq 5$)

b) Regression

$$\begin{aligned}\text{MoodScore} &= 0.25 \times 6 + 0.20 \times 4 - 0.15 \times 3.2 - 0.30 \times 2.8 + 0.10 \times 4 + 0.10 \times 3 + 2 \\ &= 1.50 + 0.80 - 0.48 - 0.84 + 0.40 + 0.30 + 2.00 \\ &= \mathbf{3.68}\end{aligned}$$

c) Final Category

$$\text{MoodScore} = 3.68 \rightarrow \leq 4 \rightarrow \mathbf{\text{Mood} = \text{Low}}$$

7. Limitations & Ethical Considerations

- **Not Clinical:** This tool provides only a general estimate, not a medical diagnosis.
- **Session-Based Data:** No inputs are stored beyond the active session; anonymity is maintained.
- **Transparency:** Users see which inputs affect their score, and thresholds are displayed in the dashboard.
- **Future Work:** We plan to refine coefficients using real survey data and cross-validation.

8. References & Further Reading

1) Sleep (S)

CDC (2023): *Sleep Deprivation, Sleep Disorders, and Chronic Disease* – highlights the strong connection between poor sleep and mental health issues like anxiety and depression [WHO Apps+15CDC+15World Health Organization+15](#).

2) Physical Activity (A)

WHO (2021): *The benefits of physical activity for mental and physical health* – summarises how regular exercise reduces symptoms of depression and anxiety [CNTW NHS](#).

3) Psychological Distress (K10)

HealthStats NSW / DHI: *Kessler 10 Psychological Distress Scale (K10)* – explains how it measures distress over the last four weeks and is widely used in population surveys [WHO Apps+10HealthStats NSW+10Cogniverse+10NovoPsych+8DHI Health+8HealthStats NSW+8](#).

4) Self-Rated Health (H)

Scopus/SAGE journal: *Self-rated mental health represents an inclusive predictor of health outcomes*, showing its predictive value for wellbeing and mortality [Social Creatures+15journals.sagepub.com+15sciencedirect.com+15](#).

5) Social Connectedness (C)

PLOS ONE (2024): *Social connectedness as a determinant of mental health* – long-term evidence that strong social support reduces depression and anxiety risk [couriermail.com.au+11journals.plos.org+11sciencedirect.com+11](#).

6) Age (D)

ABS (2024): *National Study of Mental Health and Wellbeing* – provides demographic breakdowns showing how mental health and distress vary across age groups aged 16–85 years [News.com.au+9Australian Bureau of Statistics+9Australian Bureau of Statistics+9](#).